Answer Key

Grade 6 Mathematics · Unit 2: Statistics and Data Displays · Answers for every item on Sheets 1 and 2 of each lesson packet. Constructed responses show a model answer; accept equivalent reasoning.

LESSON 2-1

Statistical Questions · Dot Plots and Shape

Sheet 1 · Statistical Questions

1. a. N b. S c. N d. S

2. Statistical. The answers (4, 5, 6, 7) are different, so the data vary.

3. Statistical. Each player lives a different distance away, so the answers vary.

4. It has one answer (one tree, one height). Model rewrite: "How tall is each tree in the park next to our school?"

5. Model: "How many hours did each person sleep last night?" Possible answers: 6, 8, 9 hours (the answers vary).

Sheet 2 · Dot Plots and Shape

1. Dots: 1 (one), 2 (three), 3 (four), 4 (three), 5 (one). a. 3 b. Symmetric (both sides match).

2. Dots: 2 (one), 3 (one), 4 (two), 5 (four), 6 (one). a. 4 b. 5 tickets c. Skewed left (tail toward 2)

3. a. 11 games b. 1 goal c. Between 4 and 7 (no games with 5 or 6 goals)

4. Skew is named for the tail, not where the dots pile up. The tail stretches right toward 7 (an outlier), so the plot is skewed right.

5. Skewed left. Most scores are high (peak at 9); the tail stretches left toward 5.

TEST PRACTICE

6. A – Symmetric

7. D – 15

LESSON 2-2

Building a Histogram · Describing Histogram Shape

Sheet 1 · Building a Histogram

1. 0–9: 3 (5, 8, 9); 10–19: 6 (11, 12, 14, 16, 17, 19); 20–29: 3 (21, 23, 25). Total 12.

2. Bars of height 3, 6, 3. a. 10–19 b. 9 games

3. a. 3 (22, 27, 29) b. 4 (31, 34, 36, 38) c. 3 (41, 45, 48). Total 10.

4. 24 − (5 + 9 + 3) = 24 − 17 = 7 students

5. The intervals are not equal widths (10, 5, and 15 values). Better: 0–9, 10–19, 20–29.

Sheet 2 · Describing Histogram Shape

1. a. 21 students b. 80–89 c. Skewed left d. 14 students

2. a. Symmetric b. Skewed right c. Skewed left d. Symmetric e. Skewed right f. Skewed left

3. a. 3 + 8 = 11 runners b. 6 + 2 = 8 runners

4. Model: 1, 2, 3, 6, 8 (sum 20). Any 5 heights with sum 20 whose tallest bars are on the right and that shrink toward the left.

5. No. The histogram only shows that 8 students scored from 80 to 89. It does not show the exact scores.

TEST PRACTICE

6. B – 10, 7, 4, 2, 1

7. C – 12

LESSON 2-3

Finding the Median · Median of an Even Data Set

Sheet 1 · Finding the Median

1. a. 6 b. 12 c. 30 d. 5

2. a. 9 students b. Ordered: 6, 6, 7, 7, 8, 8, 8, 9, 10; the 5th value is 8 hours.

3. Ordered: 4, 6, 7, 9, 11, 12, 15. Median = 9 miles.

4. 85 points. Ordered, the four known scores are 75, 82, 88, 90. The median is the 3rd of 5 values, and 85 is not one of them, so the fifth score must be 85 (75, 82, 85, 88, 90).

5. Any 5 different values with 12 in the middle when ordered, e.g. 3, 8, 12, 15, 20 (two values less than 12, two greater).

6. No. Ordered 4, 6, 9, 11, 100 still has 9 in the middle. Changing the greatest value does not move the middle value, so the median stays 9.

Sheet 2 · Median of an Even Data Set

1. a. 6 b. 16 c. 8.5 d. 37.5 e. 15 f. 6.5

2. a. 1 and 2 (the 5th and 6th values) b. (1 + 2) ÷ 2 = 1.5 cousins

3. Ordered: 5, 7, 8, 10, 12, 15. (8 + 10) ÷ 2 = 9 pounds.

4. 14. The two middle values are 10 and the missing value. Their sum is 12 × 2 = 24, so 24 − 10 = 14 (order 6, 10, 14, 20 checks).

5. She did not order the data first; 2 and 10 are the middle values of the list as written. Ordered: 2, 6, 8, 10, so the median is (6 + 8) ÷ 2 = 7.

TEST PRACTICE

6. A – 13.5

7. A – 28 and 32

LESSON 2-4

Building a Box Plot · Comparing Box Plots

Sheet 1 · Building a Box Plot

1. Ordered: 10, 14, 18, 22, 25, 30, 34, 40, 46. Min 10, Q1 16, Median 25, Q3 37, Max 46.

2. Whiskers from 10 to 16 and 37 to 46; box from 16 to 37; median line at 25.

Sheet 2 · Comparing Box Plots

1. a. Class B, by 5 min (30 vs. 25) b. A: 30 − 15 = 15 min; B: 45 − 20 = 25 min c. Class A (narrower box)

2. a. 30 − 12 = 18 b. 41 − 5 = 36

3. Maya: box width 110 − 95 = 15, range 125 − 80 = 45.

3. a. 64°F b. Q1 = 58°F, Q3 = 70°F c. About 1/2 of the days (the middle half)
4. Ordered: 11, 14, 18, 20, 26, 28, 32, 39 (even count: halves are 11, 14, 18, 20 and 26, 28, 32, 39). a. Q1 = (14 + 18) ÷ 2 = 16 b. Median = (20 + 26) ÷ 2 = 23 c. Q3 = (28 + 32) ÷ 2 = 30
5. With 7 values the median (8) is left out of both halves. The lower half is 2, 5, 7, so Q1 = 5.

Leon: box width 112 − 90 = 22, range 140 − 70 = 70. Maya is more consistent (smaller box width and range).

4. Group T. Its box is 20 − 8 = 12 cm wide; Group S's box is 16 − 12 = 4 cm wide. A wider box means more variation.
5. Bus A. Both have a typical time of 20 min, but Bus A's box is 22 − 18 = 4 min wide and Bus B's is 26 − 14 = 12 min wide. Bus A's times are more consistent, so it is easier to plan for.

TEST PRACTICE

6. B – Y's box is 31 − 25 = 6 wide; X's is 34 − 22 = 12 wide.
7. A – 21 − 12 = 9

LESSON 2-5

Range and IQR · Comparing Variability

Sheet 1 · Range and IQR

1. a. 21 − 6 = 15 b. 60 − 38 = 22 c. 31 − 18 = 13 d. 56 − 40 = 16 e. 90 − 68 = 22 f. 25 − 7 = 18
2. a. 30 − 6 = 24 pages b. 22 − 10 = 12 pages
3. Ordered: 28, 35, 41, 47, 52, 55, 60. a. 60 − 28 = 32 b. Median 47; Q1 = 35, Q3 = 55; IQR = 55 − 35 = 20
4. a. 14 + 29 = 43 b. 20 + 11 = 31
5. Half of the range is 32 ÷ 2 = 16, but the IQR is 20. The IQR depends on Q1 and Q3, not on the min and max.

Sheet 2 · Comparing Variability

1. a. 12 b. 6 (Set Q is more consistent)
2. Player 1: range 16 − 8 = 8, IQR 14 − 10 = 4. Player 2: range 22 − 2 = 20, IQR 18 − 6 = 12. Player 1 is more consistent (both medians are 12).
3. Ranges: A: 80 − 70 = 10; B: 95 − 55 = 40. More consistent: Group A (smaller range; both medians are 75)
4. a. Route 1: 40 − 5 = 35; Route 2: 25 − 8 = 17 b. Route 1: 18 − 12 = 6; Route 2: 20 − 10 = 10
5. Kwame. Route 2's range is smaller (17 vs. 35), but Route 1's range is large only because of one extreme ride (40 min). Route 1's middle half is closer together (IQR 6 vs. 10), and the IQR is the better measure when there is an extreme value.

TEST PRACTICE

6. A – Class A (smaller range)
7. A – Q1 = 6, Q3 = 12, IQR = 6

LESSON 2-6

Median and Outliers · Comparing Distributions

Sheet 1 · Median and Outliers

1. a. median 11 (outlier 48) b. median 72 (outlier 3) c. median 30.5 (outlier 95) d. median 44 (outlier 150) e. median 8 (outlier 34) f. median 54.5 (outlier 9)
2. a. 85 b. 25 min c. 8 students d. The median (25 min); the outlier 85 pulls the mean up, so 8 of 10 students are below it.
3. a. 61 (ordered: 4, 58, 59, 61, 63, 64, 66) b. (61 + 63) ÷ 2 = 62. The median changed by only 1.
4. a. $140 ÷ 7 = $20 b. $14 c. the median; $60 is an outlier, and the mean ($20) is greater than 6 of the 7 amounts
5. No. The median is 36, which sits right in the middle of the main group. Without 210 the median would be (34 + 36) ÷ 2 = 35, so the outlier changes the median by only 1. Here the median changes by only 1; the mean changes much more.

Sheet 2 · Comparing Distributions

1. a. A: 25 min, B: 35 min b. A: 30 − 20 = 10 min, B: 45 − 25 = 20 min c. Class B (greater median) d. Class A (smaller IQR)
2. a. Route 3: median 6, IQR 9 − 3 = 6; Route 8: median 6, IQR 14 − 1 = 13 b. Route 3: same median, but a smaller IQR (6 vs 13), so its times are more consistent.
3. a. Sun: median (15 + 16) ÷ 2 = 15.5 cm; Q1 = 12.5, Q3 = 18.5, IQR = 6 cm b. Shade: median (8 + 9) ÷ 2 = 8.5 cm; Q1 = 7.5, Q3 = 10, IQR = 2.5 cm. Sun plants are usually taller; shade plants are more consistent.
4. the same spread (IQR 8 min for both); Shop K's typical delivery is 8 minutes faster (median 28 vs 36).
5. Set E does have the higher center (70 > 64), but Set F is more consistent because its IQR is smaller (6 < 25). A greater median does not mean less spread.

TEST PRACTICE

6. A – smaller IQR (3 cm); the medians are equal, so center does not decide it.
7. A – Q1 = 5, Q3 = 12, IQR = 7

LESSON 2-7

Dividing a Decimal by a Whole Number · Dividing by a Decimal

Sheet 1 · Dividing a Decimal by a Whole Number

1. a. 3.2 b. 3.7 c. 0.42 d. 0.07 e. 5.2 f. 1.55
2. $16.80 ÷ 12 = $1.40 per pound (check: 1.40 × 12 = 16.80)

Sheet 2 · Dividing by a Decimal

1. a. 48 ÷ 6 = 8 b. 27.5 ÷ 5 = 5.5 c. 96 ÷ 12 = 8 d. 90 ÷ 3 = 30 e. 168 ÷ 4 = 42 f. 132 ÷ 11 = 12
2. 7.5 ÷ 1.25 = 750 ÷ 125 = 6 pillows

3. 10.2 ÷ 8 = 1.275 km (annex two zeros: 10.200)
4. a. $3.56 ÷ 4 = $0.89 b. $5.10 ÷ 6 = $0.85 c. Store B (4 cents less per can)
5. 4.35 ÷ 5 must be less than 1 because 4.35 < 5. Marcus put the decimal point in the wrong place (the point goes straight above the point in 4.35). Correct quotient: 0.87.

3. a. 2.4 ÷ 0.3 = 24 ÷ 3 = 8 glasses b. 2.4 ÷ 0.2 = 24 ÷ 2 = 12 glasses (smaller glasses, so more of them)
4. a. 4.5 ÷ 0.75 = 450 ÷ 75 = 6 laps b. 6 ÷ 0.75 = 600 ÷ 75 = 8 laps
5. She moved the divisor's point two places (0.08 to 8) but the dividend's point only one place (2.4 to 24). Both must move two places: 240 ÷ 8 = 30.

TEST PRACTICE

6. B – 67.2 ÷ 8 = 8.4
7. C – 156 ÷ 4 = 39 bags

LESSON 2-8

Finding the Mean · Reaching a Target Mean

Sheet 1 · Finding the Mean

1. a. 9 b. 25 c. 5 d. 5 e. 3.5 f. 22
2. a. 25 goals b. 10 games c. 25 ÷ 10 = 2.5 goals per game
3. Sum = 3.0 in; 3.0 ÷ 5 = 0.6 in
4. It goes up: 20 is greater than the old mean. New mean = (8 + 10 + 12 + 20) ÷ 4 = 50 ÷ 4 = 12.5
5. Zero is a data value, so there are 4 values, not 3. Mean = 24 ÷ 4 = 6 (not 8).

Sheet 2 · Reaching a Target Mean

1. A: total 36, missing 11 B: mean 16, missing 15
2. 110 × 5 = 550; 112 + 98 + 105 + 121 = 436; 550 − 436 = 114 points
3. $50 × 5 = $250; $45 + $60 + $52 + $38 = $195; $250 − $195 = $55
4. a. 91 ÷ 7 = 13 push-ups b. 14 × 8 = 112; 112 − 91 = 21 push-ups
5. She must multiply by 4, the number of tests including the new one: 85 × 4 = 340. 340 − 246 = 94, so she needs a 94.

TEST PRACTICE

6. B – 9 × 4 = 36; 36 − 25 = 11
7. C – 14 × 5 = 70

LESSON 2-9

Finding Mean Absolute Deviation (MAD) · Comparing Data Sets Using MAD

Sheet 1 · Finding Mean Absolute Deviation (MAD)

1. Distances: 5, 3, 0, 2, 6. Sum = 16. MAD = 16 ÷ 5 = 3.2
2. a. (3 + 1 + 1 + 3) ÷ 4 = 2 b. (10 + 5 + 5 + 10) ÷ 4 = 7.5 c. 0 (every value equals the mean) d. (5 + 2 + 1 + 2 + 4) ÷ 5 = 2.8
3. a. 16 ÷ 8 = 2 pets b. distances 2, 1, 1, 0, 0, 1, 1, 2; sum 8; MAD = 8 ÷ 8 = 1 pet
4. a. 96 ÷ 6 = 16 min b. distances 6, 2, 1, 1, 2, 6; sum 18; MAD = 18 ÷ 6 = 3 min (a typical wait is about 3 minutes from 16)
5. He divided by the mean (8) instead of the number of values (5). MAD = 16 ÷ 5 = 3.2

Sheet 2 · Comparing Data Sets Using MAD

1. a. (2 + 1 + 0 + 1 + 2) ÷ 5 = 1.2 points b. (8 + 4 + 0 + 4 + 8) ÷ 5 = 4.8 points c. Kai (smaller MAD)
2. a. 48 ÷ 8 = 6 books b. distances 4, 3, 2, 0, 0, 2, 3, 4; sum 18; MAD = 18 ÷ 8 = 2.25 books c. Club B (MAD 2.25 versus 0.5)
3. a. distances 2, 1, 1, 2; MAD = 6 ÷ 4 = 1.5 tomatoes b. distances 8, 4, 4, 8; MAD = 24 ÷ 4 = 6 tomatoes c. Garden A (1.5 < 6)
4. Route 5: its smaller MAD means its times stay closer to 24 minutes.
5. Set R: mean 50, distances 1, 0, 0, 1, MAD 0.5. Set S: mean 10, distances 8, 4, 4, 8, MAD 6. Jordan is wrong: bigger values do not mean more spread; Set S is more spread out.

TEST PRACTICE

6. A – the smaller MAD (2.5) means A's values stay closer to 40.
7. B – mean 6; distances 3, 1, 1, 3; 8 ÷ 4 = 2

Sheet 1 · Mean or Median?

1. a. mean (symmetric) b. median (3 is an outlier) c. median (75 is an outlier)

2. a. symmetric b. 63 ÷ 9 = 7 c. 7 d. the mean (symmetric, no outliers; mean and median are equal)

3. a. skewed right b. the mean (the few high prices in the tail pull it up) c. the median

4. a. \$78 ÷ 6 = \$13 b. (\$8 + \$8) ÷ 2 = \$8 c. the median; \$41 is an outlier, and the mean (\$13) is more than 5 of the 6 allowances

5. \$62,000 is the mean, pulled up by the one \$178,000 salary; 4 of the 5 workers earn far less. The median, \$34,000, describes typical pay better.

Sheet 2 · Matching Center and Spread

1. a. mean and MAD (symmetric) b. median and IQR (95 is an outlier)

2. a. mean and MAD (symmetric) b. mean = 32 ÷ 8 = 4 books c. distances 2, 1, 1, 0, 0, 1, 1, 2; MAD = 8 ÷ 8 = 1 book

3. a. skewed right (long right whisker) b. 14 min c. 20 − 12 = 8 min

4. a. median and IQR (70 is an outlier) b. 18 min c. Q1 = 15, Q3 = 22, IQR = 7 min

5. The value 2 is a low outlier, far below the rest, so the data is skewed left. He should report the median and IQR: median 22, IQR 24 − 20 = 4.

TEST PRACTICE

6. B – 80 is an outlier

7. A – the data is symmetric